

A PROOF THAT $s_2(3) = 13$, A VALUE ASSERTED WITHOUT PROOF BY SUN

ABSTRACT. For integers $n > 1$ and $r \geq 1$ let $s_r(n)$ be the least k such that any k vectors $a_1, \dots, a_k \in \mathbb{Z}^r$, none congruent to 0 modulo n , admit an index set I with $|I| = n$ and $\sum_{i \in I} a_i \equiv 0 \pmod{n}$ but $\sum_{i \in I} a_i \not\equiv 0 \pmod{n^2}$. Zhi-Wei Sun's Conjecture 1.2 states that $s_1(n) = 2n + 1$ and $s_2(n) = 4n + 1$ for every $n > 2$ — the s_1 half being the conjecture of Gao, Jiang, Lei, Lin and Yang — and he proved $3 \cdot 2^r + 1 \leq s_r(3) \leq 2 \cdot 3^r - 1$, i.e. $13 \leq s_2(3) \leq 17$. We prove

$$s_2(3) = 13.$$

The value is not new: Sun asserts $s_2(3) = 13$ in his Remark 1.5 and calls it easy to see, giving no proof there or elsewhere in that paper, so what is supplied here is a proof of an author-asserted value. The lower bound is Sun's. For the upper bound, Sun's own two lemmas reduce the problem to the mod-3 support T of a candidate extremal multiset and give $|S| \leq 3|T| - |A|$, and he stops at a double count that yields 16; a case analysis on $|T| \in \{0, \dots, 8\}$ carries 16 down to 12. Seven of the nine cases are immediate; the two that are not fall to a linear elimination modulo 9. The argument occupies about a page. Conjecture 1.2 remains open: $n \geq 4$ and $r \geq 3$ are untouched and the method does not generalise in either.

1. THE CONJECTURE, AND WHAT IS PROVED

Throughout, congruences between vectors are coordinatewise, and $a \not\equiv 0 \pmod{m}$ means that *some* coordinate of a is nonzero modulo m .

We quote from the e-print of Zhi-Wei Sun, *On zero-sum problems of new types*, arXiv:2606.18234 [3], version v5. Our record of the printed numbering of the defining statement is not certain — it appears as both “Definition 1.2” and “Definition 1.3” — so it is quoted by content below; “Conjecture 1.2”, “Lemma 2.1” and “Remark 1.5” are the printed numbers as read.

The definition, part (i): for $n > 1$ and $r \geq 1$, $s_r(n)$ is the least $k \in \mathbb{Z}^+$ such that any k vectors $a_1, \dots, a_k \in \mathbb{Z}^r$ with $a_i \not\equiv 0 \pmod{n}$ for all i admit a set $I \subseteq \{1, \dots, k\}$ with

$$(1) \quad |I| = n, \quad \sum_{i \in I} a_i \equiv 0 \pmod{n}, \quad \sum_{i \in I} a_i \not\equiv 0 \pmod{n^2}.$$

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Two clauses of (1) are load-bearing and are used exactly as printed: $|I| = n$ *exactly*, not $|I| \leq n$; and the indices i are distinct while the *values* a_i may repeat, so the input is a multiset.

The conjecture (Conjecture 1.2) reads: *Let $n > 2$ be an integer. Then*

$$(2) \quad s_1(n) = 2n + 1 \quad \text{and} \quad s_2(n) = 4n + 1 \quad (n > 2).$$

At $n = 3$ the second half of (2) asserts $s_2(3) = 4 \cdot 3 + 1 = 13$.

What the source proves is a bracket, not a value:

$$(3) \quad 3 \cdot 2^r + 1 \leq s_r(3) \leq 2 \cdot 3^r - 1,$$

which at $r = 2$ is $13 \leq s_2(3) \leq 17$.

What the source asserts without proof is Remark 1.5: “*It is easy to see that $s_r(3) = 2 \times 3^r + 1$ for $r = 1, 2$ but $s_3(3) > 3 \times 2^3 + 1 = 25$.*” As printed the formula gives $s_2(3) = 2 \cdot 3^2 + 1 = 19$, which the same paper’s own (3) refutes; the intended formula is $3 \cdot 2^r + 1$, forced both by the remark’s own contrast with $3 \cdot 2^3 + 1 = 25$ and by the fact that the two candidate formulas agree at $r = 1$ (both give 7) and differ only at $r = 2$. So Remark 1.5 asserts the value $s_2(3) = 13$, calls it easy, gives no proof, and is followed a few lines later by the same value being posed as an open conjecture. No proof of any exact value of $s_2(3)$ occurs in the source.

Theorem 1. $s_2(3) = 13$. *Equivalently, the largest multiset of elements of $(\mathbb{Z}/9)^2$, none congruent to 0 modulo 3, containing no three distinct indices whose sum is $\equiv 0 \pmod{3}$ and $\not\equiv 0 \pmod{9}$, has exactly 12 elements. Consequently the $n = 3$ cell of the s_2 half of (2) is true, and Sun’s interval $[13, 17]$ collapses at its upper end.*

Attribution, and how little is new. The value $s_2(3) = 13$ is asserted, without proof and as something easy to see, in Sun’s Remark 1.5. The lower bound $s_2(3) \geq 13$ is Sun’s (3); we reproduce his witness in Section 5 only so that the reader needs nothing else. Facts 3 and 4 below are Sun’s Lemma 2.1 and Sun’s own collapsing claim; together with the elementary Fact 2 they give the master bound $|S| \leq 3|T| - |A|$ of Section 2. Sun double-counts that bound to $2(3^r - 1) = 16$, which is exactly how (3) reaches 17. The only new content of this paper is Section 4, which replaces the double count by a case analysis on $|T|$ and carries 16 to 12. It is a page of elementary work, consistent with the source’s own “easy to see”.

Nearest published work. The s_1 half of (2) is the conjecture of Gao, Jiang, Lei, Lin and Yang [1], who work with integers rather than with vectors and give an upper bound of the form $3p - 2$ for primes $p \geq 3$; they say nothing about s_2 , about $\mathbb{Z}^r$ for $r \geq 2$, or about the value 13. Gao, Jiang and Mu [2] study analogous local invariants, again for integers rather than vectors, and for selections of size *at most* n rather than exactly n ; the rank-2 case and the exact-size clause of (1) do not occur there. Apart from Sun’s unproved Remark 1.5 we found no statement of the value $s_2(3)$ and no improvement

of the upper bound 17; that records what our search turned up and is not a claim of novelty.

Status: what is *not* settled. Conjecture 1.2 remains open. Only one cell of one half is proved. The case $n \geq 4$ is untouched, and the argument does not generalise in n : it uses the coincidence that the required index-set size $n = 3$ equals the number of points on a line of the affine plane $\text{AG}(2, 3)$, so Facts 2 and 4 have no analogue for $n \neq 3$. The case $r \geq 3$ is untouched, and the source itself records that the pattern breaks there ($s_3(3) > 25$). The s_1 half of (2) is not addressed here. No claim of minimality or uniqueness is made for the extremal multisets: we exhibit one multiset of size 12 and do not classify them. Finally, the correction of Remark 1.5's misprint is not offered as a separate result: it follows from the source's own (3) with no new mathematics.

2. REDUCTION, AND THE THREE FACTS

Every condition in (1) at $n = 3$ depends only on residues modulo $n^2 = 9$, so we work in $(\mathbb{Z}/9)^2$. Call $a \in (\mathbb{Z}/9)^2$ *admissible* if $a \not\equiv 0 \pmod{3}$; there are $9^2 - 3^2 = 72$ admissible classes. Call a multiset S of admissible classes *bad* if no three distinct indices of S have a sum that is $\equiv 0 \pmod{3}$ and $\not\equiv 0 \pmod{9}$. Being bad is inherited by sub-multisets, so

$$(4) \quad s_2(3) = 1 + \max\{|S| : S \text{ bad}\}.$$

Theorem 1 is therefore the statement $\max |S| = 12$.

Write $\pi: (\mathbb{Z}/9)^2 \rightarrow \mathbb{F}_3^2$ for reduction modulo 3. For $v \in \mathbb{F}_3^2 \setminus \{0\}$ let N_v be the sub-multiset of S with $\pi = v$, and let $T = \{v : N_v \neq \emptyset\}$, the *support*, so $|S| = \sum_{v \in T} |N_v|$. Call a 3-sub-multiset of S a *candidate* if its reductions sum to 0 in $\mathbb{F}_3^2$; then S is bad if and only if every candidate of S sums to 0 modulo 9.

Fact 2 (candidate shapes; elementary). *If $v_1 + v_2 + v_3 = 0$ in $\mathbb{F}_3^2$ with every $v_i \neq 0$, then either $v_1 = v_2 = v_3$, or v_1, v_2, v_3 are distinct and are the three points of an affine line of $\text{AG}(2, 3)$ not through the origin. There are exactly 8 such lines; each of the 8 nonzero points lies on exactly 3 of them; and no such line contains an antipodal pair $\{v, -v\}$, so each meets exactly 3 of the 4 antipodal pairs, in one point each.*

Proof. If $v_1 = v_2 \neq v_3$ then $v_3 = -2v_1 = v_1$ because $-2 \equiv 1 \pmod{3}$, a contradiction; so the v_i are all equal or all distinct, and in the latter case $v_3 = -v_1 - v_2$ is the third point of the line through v_1 and v_2 . $\text{AG}(2, 3)$ has 12 lines, of which 4 pass through 0, leaving 8; each of the 8 nonzero points lies on 4 lines, one of them through 0, hence on 3 of the 8, and $8 \cdot 3 = 24 = 8 \cdot 3$ counts incidences both ways. If a line missing 0 contained v and $-v$ its third point would be $-v - (-v) = 0$. $\square$

Fact 3 (Sun's Lemma 2.1, source line 736). *For a bad S : (a) no admissible class occurs three times in S ; and (b) $|N_v| \leq 3$ for every v .*

Proof. Three elements of one N_v are a candidate, since $3v = 0$, so they must sum to 0 modulo 9. (a) Three copies of a sum to $3a$, which is $\equiv 0 \pmod{3}$ and is $\equiv 0 \pmod{9}$ only if $a \equiv 0 \pmod{3}$, excluded. (b) If $a, b, c, d \in N_v$ are four elements then $a + b + c = a + b + d = 0$ gives $c = d$ and $a + c + d = b + c + d = 0$ gives $a = b$, so $|N_v| \geq 4$ forces four copies of one class, contradicting (a). $\square$

Fact 4 (Sun's claim, source lines 741–760). *Call a line L of $\text{AG}(2, 3)$ missing the origin full (for S) if all three of its points lie in T . If $L = \{v_1, v_2, v_3\}$ is full, then each of $N_{v_1}, N_{v_2}, N_{v_3}$ consists of one class repeated, and that class triple sums to 0 modulo 9; in particular $|N_{v_i}| \leq 2$ for $i = 1, 2, 3$ by Fact 3(a).*

Proof. Any $a \in N_{v_1}, b \in N_{v_2}, c \in N_{v_3}$ form a candidate, so $a + b + c = 0$ modulo 9; for $a, a' \in N_{v_1}$ this gives $a = a'$. $\square$

Let $A = A(T)$ be the union of the point sets of all full lines. Since $|N_v| \leq 2$ for $v \in A$ and $|N_v| \leq 3$ otherwise, Facts 3 and 4 give the

$$(5) \quad \text{master bound:} \quad |S| \leq 3|T| - |A(T)|.$$

This is where the source stops: double counting over the origin-missing lines turns (5) into $\max |S| \leq 2(3^r - 1)$, which is 16 at $r = 2$ and gives the published $s_2(3) \leq 17$ of (3).

3. COORDINATES

The four antipodal pairs of $\mathbb{F}_3^2 \setminus \{0\}$ are

$$\begin{aligned} P_1 &= \{(1, 0), (2, 0)\}, & P_2 &= \{(0, 1), (0, 2)\}, \\ P_3 &= \{(1, 1), (2, 2)\}, & P_4 &= \{(1, 2), (2, 1)\}, \end{aligned}$$

the first member of each being called $+$. By Fact 2 there are $\binom{4}{3} = 4$ triples of pairs and two lines per triple (a line and its negative), 8 in all. Name the frozen classes of Fact 4 by

$$\begin{aligned} A &= x_{(1,0)}, & A' &= x_{(2,0)}, & B &= x_{(0,1)}, & B' &= x_{(0,2)}, \\ C &= x_{(1,1)}, & C' &= x_{(2,2)}, & D &= x_{(1,2)}, & D' &= x_{(2,1)}, \end{aligned}$$

each an element of $(\mathbb{Z}/9)^2$ reducing to its subscript. Then the 8 lines and the 8 equations they impose, all modulo 9, are:

	line	equation	pairs met
E_1	$(1, 0) (0, 1) (2, 2)$	$A + B + C' = 0$	$P_1 P_2 P_3$
E_2	$(1, 0) (0, 2) (2, 1)$	$A + B' + D' = 0$	$P_1 P_2 P_4$
E_3	$(1, 0) (1, 1) (1, 2)$	$A + C + D = 0$	$P_1 P_3 P_4$
E_4	$(2, 0) (0, 2) (1, 1)$	$A' + B' + C = 0$	$P_1 P_2 P_3$
E_5	$(2, 0) (0, 1) (1, 2)$	$A' + B + D = 0$	$P_1 P_2 P_4$
E_6	$(2, 0) (2, 2) (2, 1)$	$A' + C' + D' = 0$	$P_1 P_3 P_4$
E_7	$(0, 1) (1, 1) (2, 1)$	$B + C + D' = 0$	$P_2 P_3 P_4$
E_8	$(0, 2) (2, 2) (1, 2)$	$B' + C' + D = 0$	$P_2 P_3 P_4$

TABLE 1. The eight lines of $\text{AG}(2, 3)$ missing the origin (E_3, E_6 are $x = 1, x = 2$ and E_7, E_8 are $y = 1, y = 2$).

Finally, a relabelling remark used once. A matrix with entries in $\{0, 1, 2\}$ and determinant nonzero modulo 3 has determinant coprime to 3, hence a unit modulo 9; so $GL_2(\mathbb{F}_3)$ lifts into $GL_2(\mathbb{Z}/9)$ and acts on $(\mathbb{Z}/9)^2$ preserving admissibility, “ $\equiv 0 \pmod{3}$ ”, “ $\equiv 0 \pmod{9}$ ” and all incidences. Its action on $\mathbb{F}_3^2 \setminus \{0\}$ is transitive, and its action on $\{P_1, P_2, P_3, P_4\}$ is the full S_4 (the kernel is $\{\pm I\}$ and $48/2 = 24$).

4. THE UPPER BOUND: $\max |S| \leq 12$

The support size $|T|$ runs over $\{0, 1, \dots, 8\}$, which partitions every conceivable bad S ; we take the nine values in turn. Throughout, $|S| \leq 3|T| - |A(T)|$ is (5).

$|T| \leq 4$. Then $|S| \leq 3|T| \leq 12$, and nothing else is needed.

$|T| = 5$. The distribution of T over the four antipodal pairs is $(2, 2, 1, 0)$ or $(2, 1, 1, 1)$ up to order.

$(2, 2, 1, 0)$: two pairs $P_i, P_j \subseteq T$, and P_k contributes a single point of sign s . The triple (P_i, P_j, P_k) carries two lines whose P_k -signs are opposite, so one of them has P_k -sign s while its other two points lie in $P_i \cup P_j \subseteq T$. That line is full, so $|A| \geq 3$ and $|S| \leq 15 - 3 = 12$.

$(2, 1, 1, 1)$: relabel so that $P_1 \subseteq T$ and T contains $P_2^{s_2}, P_3^{s_3}, P_4^{s_4}$. Reading Table 1, a line on a given triple of pairs is full exactly when: on (P_1, P_2, P_3) , $s_2 \neq s_3$; on (P_1, P_2, P_4) , $s_2 = s_4$; on (P_1, P_3, P_4) , $s_3 = s_4$; on (P_2, P_3, P_4) , $s_2 = s_3 \neq s_4$. If $s_2 \neq s_3$ then s_4 agrees with one of them, so the second or third condition holds; if $s_2 = s_3 = s_4$ the second holds; if $s_2 = s_3 \neq s_4$ the fourth holds. All 8 sign vectors are covered, a full line always exists, and $|S| \leq 12$.

$|T| = 6$. The distribution is $(2, 2, 2, 0)$ or $(2, 2, 1, 1)$, and in both cases $A(T) = T$, so $|S| \leq 18 - 6 = 12$.

$(2, 2, 2, 0)$: relabel $P_1, P_2, P_3 \subseteq T$, $P_4 \cap T = \emptyset$. Both lines on (P_1, P_2, P_3) are full and their union is all six points of $P_1 \cup P_2 \cup P_3 = T$.

$(2, 2, 1, 1)$: relabel $P_1, P_2 \subseteq T$ together with $P_3^{s_3}$ and $P_4^{s_4}$. Since $-I$ flips all four signs at once, (s_3, s_4) has two orbits. For $(-, -)$ the three full lines $\{P_1^+, P_2^+, P_3^-\}$, $\{P_1^+, P_2^-, P_4^-\}$, $\{P_1^-, P_3^-, P_4^-\}$ have union T ; for $(-, +)$ the three full lines $\{P_1^+, P_2^+, P_3^-\}$, $\{P_1^-, P_2^+, P_4^+\}$, $\{P_2^-, P_3^-, P_4^+\}$ have union T .

The two remaining cases are the ones (5) does not settle: it gives $3 \cdot 7 - 7 = 14$ and $3 \cdot 8 - 8 = 16$. Both are excluded by one linear elimination modulo 9 in the notation of Table 1.

Proposition 5. *No bad S has $|T| \geq 7$.*

Proof. It suffices to treat $|T| = 7$: if S is bad with $|T| = 8$, then deleting one whole fibre N_v leaves a bad multiset with support of size 7.

By transitivity of $GL_2(\mathbb{F}_3)$ on $\mathbb{F}_3^2 \setminus \{0\}$ we may relabel the missing point to $(2, 1)$. The full lines are exactly those rows of Table 1 that avoid $(2, 1)$,

namely E_1, E_3, E_4, E_5, E_8 , and their union is all seven points of T ; so by Fact 4 every fibre is a single class and those five equations all hold. Now E_1 gives $C' = -A - B$; E_3 gives $D = -A - C$; E_5 gives $A' = -B - D = A - B + C$; E_4 gives $B' = -A' - C = -A + B - 2C$; and substituting into E_8 ,

$0 = B' + C' + D = (-A + B - 2C) + (-A - B) + (-A - C) = -3(A + C)$ modulo 9, so $A + C \equiv 0 \pmod{3}$. But $\pi(A) + \pi(C) = (1, 0) + (1, 1) = (2, 1) \neq 0$, a contradiction. $\square$

Combining: $|T| \leq 6$ gives $|S| \leq 12$ by (5), and $|T| \geq 7$ is impossible. Hence $\max |S| \leq 12$ and, by (4), $s_2(3) \leq 13$.

5. THE LOWER BOUND IS SUN'S, WITH HIS WITNESS

The inequality $s_2(3) \geq 13$ is the left half of (3) and is not ours. For completeness — so that a reader needs nothing beyond this page — here is a multiset attaining it. Let

$$(6) \quad W = \{ (1, 0), (4, 0), (4, 0), (8, 0), (5, 0), (5, 0), \\ (0, 1), (0, 4), (0, 4), (0, 8), (0, 5), (0, 5) \}$$

in $(\mathbb{Z}/9)^2$. This is $4 \cdot \{1, 1, 8, 8, 2, 7\} \pmod{9}$, i.e. the unit multiple 4 of Sun's own one-dimensional extremal set ($n - 1$ copies of $+1$, $n - 1$ of -1 , and $\pm(n - 1)$), placed on each coordinate axis.

Lemma 6. *W is a bad multiset of 12 admissible classes. Hence $\max |S| = 12$ and, with Section 4 and (4), $s_2(3) = 13$.*

Proof. Each of the twelve entries is nonzero modulo 3, and $|W| = 12$. The reductions are $(1, 0)$ three times, $(2, 0)$ three times, $(0, 1)$ three times and $(0, 2)$ three times, so $T = P_1 \cup P_2$ has size 4, every fibre has size 3, and no class occurs more than twice. Since T consists of two whole antipodal pairs, Fact 2 says T contains no origin-missing line, so $A(T) = \emptyset$ and every candidate of W is a same-fibre triple. There are exactly four of them, one per fibre, and

$$(1, 0) + (4, 0) + (4, 0) = (9, 0), \quad (8, 0) + (5, 0) + (5, 0) = (18, 0),$$

$$(0, 1) + (0, 4) + (0, 4) = (0, 9), \quad (0, 8) + (0, 5) + (0, 5) = (0, 18),$$

all $\equiv 0 \pmod{9}$. So W is bad, giving $s_2(3) \geq 13$. $\square$

6. VERIFICATION

Nothing above needs a computer. The finite claims are nevertheless machine-checked from the objects printed here — the multiset (6) and Table 1 — by `verify.py`, a standard-library Python program accompanying this note, in exact arithmetic modulo 3 and modulo 9. It checks that there are 72 admissible classes; that the candidate reduction-triples split into 8 all-equal and 8 all-distinct with none mixed, that the all-distinct ones are exactly the 8 origin-missing lines with the incidence and antipodal counts of Fact 2, and

that the rows of Table 1 are those lines with the equations printed; that $3a \not\equiv 0 \pmod{9}$ for all 72 admissible a , the mechanism of Fact 3(a); that W is admissible and bad, of size 12, with the support, candidate list and candidate sums stated above; that $\max(3|T| - |A(T)|)$ over all $2^8 = 256$ supports is 0, 3, 6, 9, 12, 12, 12, 14, 16 for $|T| = 0, \dots, 8$, so that (5) settles 247 supports and leaves exactly the $\binom{8}{7} + \binom{8}{8} = 9$ with $|T| \geq 7$; that the elimination displayed in Proposition 5 is an identity over all $9^3 = 729$ lifts of (A, B, C) ; and — going beyond what Section 4 argues — that each of the eight supports of size 7 and the one of size 8 is inconsistent modulo 9 on its own, so that the exclusion does not depend on the relabelling. For the last of these, the substitution $x_v = v + 3u_v$ with $u_v \in \mathbb{F}_3^2$ turns each full-line equation $x_p + x_q + x_r \equiv 0 \pmod{9}$ into a linear equation $u_p + u_q + u_r = -w_L$ over $\mathbb{F}_3$, and those systems are solved exhaustively. The recorded run, filed alongside the program as `verify.output.txt`, reports 64 checks, all passing.

The program does not re-derive the cancellation steps of Fact 3(b) and Fact 4 or the counting in (5), which are proofs, and it performs no independent exhaustive search over multisets.

REFERENCES

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- [3] Zhi-Wei Sun, *On zero-sum problems of new types*, arXiv:2606.18234, 2026. arXiv:2606.18234. We read version v5 of the e-print; a printed version, if any, may number its statements differently.